

Constraint encodings, not annealers, dominate QUBO/HUBO siting for urban heat-island interventions

Full experimental record

Repository `s-siyanwal/q_uhi`, branch `claude/clever-gates-ezote4`, commit `e649582`

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Abstract. We site parks, water bodies and cool pavement on synthetic 50 m city grids. Each instance is a constrained binary program, encoded as a QUBO or HUBO and scored on exact saturating cooling physics against a HiGHS-certified optimum. A slack-bit penalty QUBO with a corrected weight has the MILP optimum as its ground state on all 11 exhaustively enumerated instances. The second-order Taylor QUBO loses 10.3% of the achievable cooling at 300 m reach, where the cubic HUBO loses at most 0.02% and a least-squares-fitted QUBO at most 0.42%. The final annealing gap scales as $1/\Lambda$ with the penalty weight. Constraint-preserving Dicke-XY QAOA reaches a mean $P(\text{opt})$ of 0.43 at depth 6, against 1.00 for feasible-space simulated annealing. Path-integral Monte Carlo restricted to the feasible set passes a pre-committed comparability bar (benefit 0.981), but it made $3.6\times$ more proposals than feasible-space annealing; at equal proposals it scores 0.983 against 0.988. HiGHS certifies every hard-family instance in 0.77 s on average. Encodings and constraint handling, not the choice of annealer, decide the outcome. We observe no quantum or quantum-inspired advantage.

Success criterion (fixed in `docs/NEXT.md` before the quantum experiments). A quantum or quantum-inspired method is “at par” only if, at matched wall-clock on certified instances, its benefit ratio (exact physics) and $P(\text{opt})$ meet the *greedy planner* and *HiGHS MILP* on *both* the park and the mix + equity families. Matching SA or SA-matched is not enough. SQA is quantum-inspired PIMC, not hardware quantum annealing.

Verdict. Not met. Gate-model QAOA stays below the bar; feasible-space PIMC ties feasible-space SA and does not beat HiGHS. No D-Wave or other hardware numbers exist in this repository.

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How to read this report

This is the exhaustive record behind the two short papers. Every number in it is copied from `docs/RESULTS.md`, `docs/AUDIT.md`, `docs/NEXT.md`, `docs/PAPER.md`, or a `results/*/summary.csv` / `e*.md` file; tables marked “generated” are produced from the committed CSVs by `latex/tools/make_tables.py`. No experiment was rerun to produce this document. Each experiment chapter has the same structure: *question*, *setup*, *output files*, a *table* from the official summary, and a *reading*. Two kinds of boxes appear:

In plain language. a two- or three-sentence summary for readers who are not optimisation specialists.

For specialists. details that matter for reproducing or judging the result (definitions, caveats, what is and is not simulated).

Chapter 1

Introduction and problem

1.1 Why this study

Quantum and quantum-inspired optimisers are often proposed for spatial planning problems that can be written as QUBOs. Urban-heat-island (UHI) mitigation is one such problem: choose where to put parks, water bodies and cool pavement on a city grid, under a budget and an equity requirement, so that the population-weighted air temperature is as low as possible. This work asks a narrow question: *on certified instances and at matched effort, does any quantum or quantum-inspired method reach the benefit of the classical baselines?*

We answer it with a clean re-implementation, the quhi toolkit, after an audit found that an earlier code base produced penalty artefacts rather than results (Section 2). Every plan is scored on the exact saturating cooling physics and compared with a HiGHS-certified optimum [2]. The success criterion was fixed in advance: a method is at par only if, at matched wall-clock on certified instances, it meets the greedy planner *and* the MILP on both instance families. Matching simulated annealing is not enough.

The answer is negative, and the reasons are specific. The encoding decides most of the outcome: slack penalties set the annealing gap, Taylor truncation distorts long-reach cooling, and quadratisation hides native higher-order structure. Once constraints are handled natively, a feasible-space classical annealer is as good as the best quantum-inspired sampler we built, and HiGHS certifies every hard instance in about a second.

In plain language. We asked whether quantum or quantum-inspired computers help decide where to put parks, water and cool pavement in a hot city. On every test we ran, a standard classical solver found the provably best plan in about a second, and the best quantum-inspired method was no better than an ordinary classical search that respects the rules.

1.2 The planning problem

Following the Sanya green-space study [6], a city is a grid of 50×50 m cells with fixed land uses. A decision variable $x_v \in \{0, 1\}$, $v = (i, o)$, places option $o \in \{\text{park, water, cool pavement}\}$ on candidate cell i . The cooling that option o at cell i delivers to target cell k is $e_{ik} = A_o \exp(-d_{ik}/L_o)$, with illustrative defaults $A = 1.5^\circ\text{C}$, $L = 100$ m (park), $A = 2.0^\circ\text{C}$, $L = 80$ m (water) and $A = 0.6^\circ\text{C}$, $L = 30$ m (cool pavement), at costs of 3, 5 and 1 units.

Saturation. Overlapping cool islands do not add without limit. We use

$$\Delta T_k(x) = \Delta T_{\max} \left[1 - \prod_v (1 - p_{kv} x_v) \right], \quad p_{kv} = \min(e_{kv}, \Delta T_{\max}) / \Delta T_{\max}, \quad (1.1)$$

with $\Delta T_{\max} = 3^\circ\text{C}$. The objective is the exposure-weighted temperature $f(x) = \sum_k w_k (T_k^0 - \Delta T_k(x))$, $w_k = 0.7 \text{pop}_k / \sum \text{pop} + 0.3/N$. Inclusion–exclusion truncated at order K gives an additive model ($K=1$), a QUBO ($K=2$) or a cubic HUBO ($K=3$). By the Bonferroni inequalities the QUBO is a pessimistic bound on the exact objective. An optional park-block synergy adds a quartic term for every complete 2×2 block of parks.

Constraints. An integer budget $\sum_v c_o x_v \leq B$; at most one option per cell; and, optionally, at least one intervention per district (equity). With one option per cell and no equity, $-f$ is monotone submodular, so the cost-benefit greedy planner has a $(1 - 1/e)$ guarantee [4]; any quantum claim must therefore beat greedy and MILP, not only SA.

Scoring. $\text{benefit}_{\text{true}}$ is the exact-physics cooling of a returned plan divided by that of the MILP plan; an infeasible plan scores 0. Because MILP certifies the $K=2$ surrogate and not Eq. (1.1), $\text{benefit}_{\text{true}}$ can slightly exceed 1. $P(\text{opt})$ is the fraction of runs (or the exact probability, for QAOA) that return the certified optimum.

For specialists. The QUBO ($K=2$) objective is a Bonferroni lower bound on the exposure temperature, so it is pessimistic; each extra order tightens the bracket. MILP certifies the *surrogate* optimum, and all benefits are measured on the *exact* law, which is why values slightly above 1 occur (for example 1.0001 for greedy on park instances in E8).

1.3 Instance families

- **Park family** (E3, E8): park only, budget constraint, no equity.
- **Mix + equity family** (E3, E8, E9): park, water and cool pavement, at least one intervention per district.
- **“Exactly 3 parks”** (E8, E8b): 7×7 park cities with a cardinality constraint, $n = 10$, $|\mathcal{F}| = 120$, no slack.
- **Family H** (E12, E12b): 8×8 , 10×10 and 12×12 cities (seeds 100–102, candidate fraction 0.3), three intervention types, budget equal to the grid side, equity; the two larger sizes add a 0.3°C park-block synergy (quartic terms).
- **Family C** (E12, E12b): three park-only 8×8 cities (seeds 100–102).
- **E13 tile**: one LST-shaped 16×16 tile (seed 0).

All cities are synthetic (`quhi.uhi.generate_city`).

Chapter 2

Legacy audit (F1–F8)

The repository started from an earlier code base and a set of design documents. The audit (docs/AUDIT.md, reproduced by scripts/audit_legacy.py) found that its results cannot be used; 19 of its 163 tests fail. The findings that matter for this paper are:

- F1 inequality constraints encoded as equalities in raw m^2 , giving a coefficient ratio of 1.72×10^9 against 2.0 and an infeasible global minimum;
- F2 shipped “results” whose energies are penalty magnitudes, with an all-zero SQA plan;
- F3 an SQA action with β instead of β/P on the problem term: the single-qubit $\langle \sigma_z \rangle$ is -0.997 against the exact -0.545 (quhi: -0.546);
- F4 a penalty theorem that is false for non-integer data (the corrected bound is about $2\times$ smaller, median ratio 2.06);
- F5 an automatic penalty weight that can be too small by a factor of n ;
- F6 a QUBO $\leftrightarrow$ Ising round trip that doubles couplings;
- F7 worked examples and “known optima” that are wrong.

The legacy energies are artefacts of F1–F3 and are not cited anywhere in this work. The legacy material also invoked asymptotic quantum-annealing results for one-dimensional disordered chains [5]; those do not transfer to dense, penalty-dominated QUBOs (E6).

Table 2.1: Legacy audit findings (from docs/AUDIT.md).

	finding	evidence (docs/AUDIT.md, results/audit/audit.json)
F1	inequalities encoded as equalities, in raw m^2	$ Q _{\max} = 1.72 \times 10^9$ vs $ Q_{\text{phys}} _{\max} = 2.0$; infeasible global minimum
F2	shipped “results” are artefacts	energies are penalty magnitudes; SQA returns the all-zero plan
F3	SQA uses β instead of β/P	$\langle \sigma_z \rangle$: exact -0.545 , quhi -0.546 , legacy -0.997
F4	penalty theorem false for non-integer data	counter-example; corrected bound about $2\times$ smaller (median 2.06)
F5	automatic penalty weight not a bound	too small by up to a factor n ($Q = 1$, $n = 10$: 10 vs 100)
F6	QUBO $\leftrightarrow$ Ising round trip doubles couplings	round trip returns doubled off-diagonals
F7	wrong worked examples and “known optima”	claimed -19.25 where $f = -5$ and the optimum is -6

F8 (minor/major), other issues. The sign of the pairwise physics is inconsistent in the plan (synergy in one place, diminishing returns in another); quhi derives it from the saturation law (pairs positive, triples negative).

Table 2.2: F3: single-qubit thermal expectation ($h = 0.5, \Gamma = 0.7, \beta = 2, P = 16$).

	$\langle \sigma_z \rangle$
exact quantum	−0.545
quhi PIMC	−0.546
legacy action	−0.997

The legacy suite has 19 failures, 141 passes and 3 skips (qiskit tests excluded); several passing tests check only shapes. QAOA/VQE modules need qiskit and are stress-tested rather than checked for correctness. Results carry timestamps but no seeds or environment details. The claimed quantum scaling rests on one-dimensional disordered-chain results [5]; E6 shows that with the certified penalty the objective’s level spacing is 7×10^{-6} of the spectrum width at 14 qubits and shrinks as $1/\Lambda$.

In plain language. The earlier code produced numbers of order 10^{10} that are penalty sizes, not temperatures, and a “quantum” sampler that was really a very cold classical one. We rebuilt the toolkit from scratch and do not use any of those numbers.

Full details and the `audit.json` pointers are in Appendix C.

Chapter 3

Toolkit surface

quhi is a Python package (numpy, scipy with HiGHS, numba, matplotlib, pandas).

module	role
<code>quhi.polynomial</code>	BinaryPolynomial: sparse pseudo-Boolean polynomial of any degree (QUBO/HUBO), upper-triangular QUBO export
<code>quhi.constraints</code>	ConstrainedBinaryProgram, LinearConstraint; bounded-slack penalty model, certified Λ_{safe} , unbalanced/linear forms
<code>quhi.quadraticize</code> <code>quhi.uhi</code>	Rosenberg substitution with a proved sufficient M synthetic cities, interventions, saturating cooling, $K = 1, 2, 3$ surrogates, fitted QUBO, park-block synergy, greedy planner
<code>quhi.solvers.exact</code> <code>quhi.solvers.classical</code> <code>quhi.solvers.quantum</code> <code>quhi.solvers.mixers</code>	HiGHS MILP, exhaustive enumeration, exact spectra and annealing SA, Tabu (numba kernels, native k -local deltas) SQA (PIMC, β/P), state-vector QAOA ConstrainedQAOA (Dicke/XY, per-cell XY, feasibility-projected), simulated on $\text{span}(\mathcal{F})$
<code>quhi.solvers.feasible*</code> <code>quhi.solvers.warmstart</code> <code>quhi.solvers.cqm</code>	FeasibleSA, FeasibleSQA, Tabu-on-F QAOA-seeded FeasibleSA (declared hybrid) CQM constraint spec, <code>to_cqm</code> , and <code>sample_cqm</code> , a Leap hybrid hook that returns “skipped” without DWAVE_API_TOKEN
<code>quhi.postprocess</code> <code>quhi.analysis</code>	repair and feasible local search metrics, benchmarking, plotting, GIF animation

For specialists. The test suite (`pytest -q`, 133 tests) checks, among other things: the penalty lemma and theorem on random programs; that too small Λ or M fails; the XY mixer against an explicit 2^n Pauli construction; subspace against full state-vector QAOA; feasibility at every FeasibleSA/FeasibleSQA step; PIMC against the exact single-qubit $\langle \sigma_z \rangle$; and that `sample_cqm` makes no network call without a token.

Chapter 4

Encodings and surrogates

4.1 E1: encoding certification

Question. Is the slack-penalty QUBO with the corrected weight exact?

Setup. Exhaustive enumeration of the full penalty QUBO (decision plus slack variables) compared with HiGHS on the constrained program; 11 instances (6×6 and 8×8 park, 6×6 mix, one with equity).

Output. `results/E1_encoding/encoding-certification.csv, summary.json`

Reproduce. `python scripts/run_experiments.py --only E1`

Table 4.1: E1 (generated).

instance	dec.	slack	MILP opt. (°C)	QUBO ground (°C)	match	Λ_{safe}	range/safe
6x6_s0_park	5	3	30.818	30.818	yes	1.906	1.881
6x6_s1_park	5	3	30.755	30.755	yes	1.825	1.839
6x6_s2_park	5	3	30.720	30.720	yes	1.845	1.854
6x6_s3_park	5	3	30.823	30.823	yes	2.045	1.815
8x8_s0_park	12	4	30.584	30.584	yes	2.961	2.124
8x8_s1_park	12	4	30.507	30.507	yes	2.330	2.208
8x8_s2_park	12	4	30.738	30.738	yes	2.417	2.193
8x8_s3_park	12	4	30.757	30.757	yes	2.706	2.146
6x6_s0_mix	15	3	30.369	30.369	yes	4.850	2.063
6x6_s1_mix	15	3	30.339	30.339	yes	4.538	2.018
8x8_s1_park + equity	12	11	30.552	30.552	yes	2.403	2.142

Reading. The ground state equals the MILP optimum on 11 of 11 instances; Λ_{safe} is $1.81\text{--}2.21\times$ smaller than the range bound (median 2.06). One configuration was skipped because its equity demand exceeds the budget. Some 6×6 instances show ground-state degeneracy 2, an artefact of the bounded slack encoding that is harmless for exactness.

In plain language. The translation of rules into penalties is correct: the lowest-energy answer is always the best legal plan.

4.2 E2: surrogate fidelity

Question. How much cooling does each polynomial surrogate lose when its optimum is scored on the exact physics?

Setup. 12×12 cities, 22 candidate cells, park budget ≤ 6 , 4 seeds per decay length; each surrogate's optimum found by enumerating all feasible plans.

Output. results/E2_fidelity/

Reproduce. python scripts/run_experiments.py --only E2

Table 4.2: E2: share of achievable cooling lost (generated).

L (m)	additive $K=1$	QUBO $K=2$ (Taylor)	QUBO (LSQ fit)	HUBO $K=3$
50	0.20\{\}\%	0.00\{\}\%	0.00\{\}\%	0.00\{\}\%
100	0.00\{\}\%	0.20\{\}\%	0.02\{\}\%	0.00\{\}\%
200	0.29\{\}\%	2.81\{\}\%	0.21\{\}\%	0.02\{\}\%
300	0.00\{\}\%	10.30\{\}\%	0.42\{\}\%	0.00\{\}\%

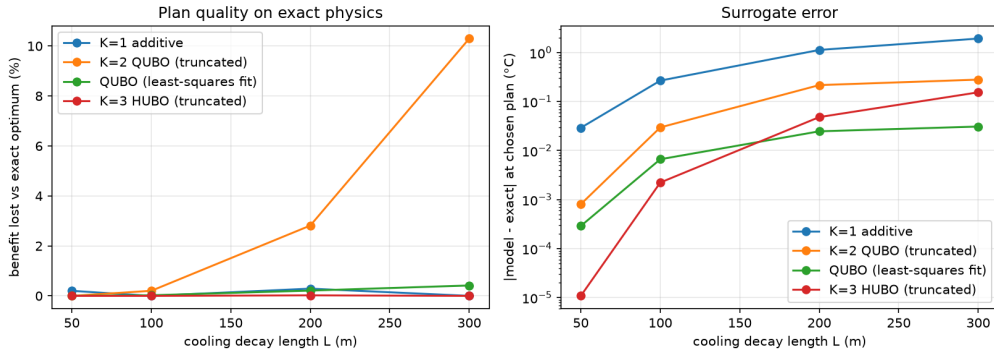


Figure 4.1: E2 fidelity versus decay length.

Reading. When kernels overlap strongly, the second-order truncation over-counts diminishing returns, believes clustered parks are much worse than they are, and spreads them out: at $L = 300$ m the Taylor QUBO loses 10.30%. The cubic HUBO loses at most 0.02%; a QUBO with the same 253 coefficients fitted by least squares loses at most 0.42%. The number of terms grows from 22 ($K=1$) to 253 ($K=2$) to 1,793 ($K=3$).

4.3 E4: penalty-weight sweep

Question. How does the penalty weight trade feasibility against sampler quality?

Setup. $\Lambda/\Lambda_{\text{safe}} \in \{0.01, \dots, 3\}$, 3 seeds $\times$ 32 reads, mix 8×8 + equity and park 12×12 .

Output. results/E4_penalty/

Reproduce. python scripts/run_experiments.py --only E4

Table 4.3: E4 (typed from docs/RESULTS.md).

instance	solver	$\Lambda/\Lambda_{\text{safe}}$	$P(\text{feas})$ raw	$P(\text{opt})$ raw	$P(\text{opt})$ pp	benefit pp
mix 8×8 + equity	SA	0.01	0.10	0.01	0.42	0.78
	SA	1	1.00	0	0.03	0.77
	SQA	0.01	0.00	0	0.29	0.94
	SQA	1	1.00	0	0.01	0.89
	Tabu	0.01	0.20	0.01	0.47	0.83
	Tabu	1	1.00	0	0	0.71
park 12×12	all	≤ 0.03	≤ 0.15	≈ 0	1.00	1.00
	Tabu	≥ 0.1	1.00	0.97–0.99	1.00	1.00

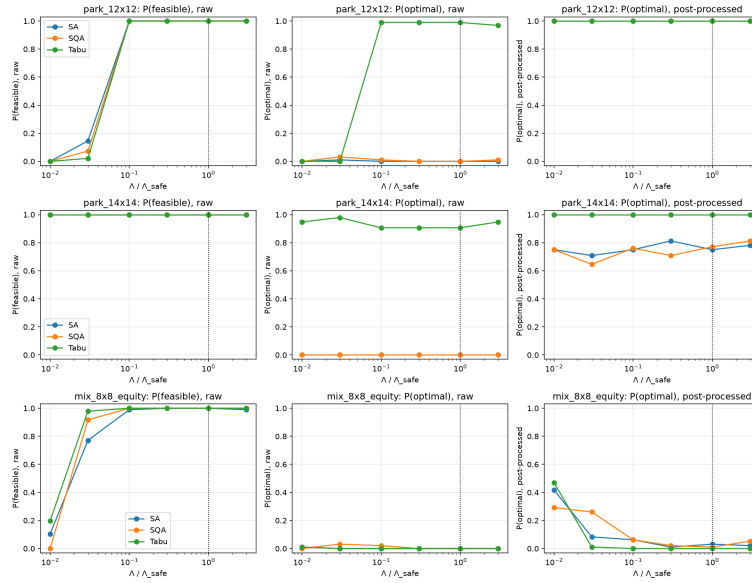


Figure 4.2: E4 penalty sweep.

Reading. Below about $0.1\Lambda_{\text{safe}}$ raw feasibility collapses, as the theory predicts, but the infeasible low-penalty samples are better starting points for repair: $P(\text{opt})$ after post-processing rises from at most 0.03 to 0.42 (SA) and 0.47 (Tabu). The recipe is hybrid and not certified.

4.4 E5: native HUBO versus quadratised QUBO

Question. Should higher-order objectives be solved natively?

Setup. 10×10 cities, 30 decision variables, cubic saturation plus quartic park-block synergy; the constrained HUBO optimum is certified by MILP on all three instances.

Output. `results/E5_hubo/`

Reproduce. `python scripts/run_experiments.py --only E5`

Table 4.4: E5 (generated from `hubo_summary.csv`).

formulation	solver	vars	P(opt) raw	pp	benefit
QUBO (Rosenberg)	Greedy	192	0.000	1.000	0.849
QUBO (Rosenberg)	SA	192	0.000	1.000	0.837
QUBO (Rosenberg)	SQA	192	0.000	0.997	0.930
QUBO (Rosenberg)	Tabu	192	0.000	1.000	0.847
native HUBO	Greedy	30	0.003	0.976	0.921
native HUBO	SA	30	0.000	0.997	0.920
native HUBO	SQA	30	0.000	0.993	0.960
native HUBO	Tabu	30	0.990	1.000	0.999

Reading. Tabu finds the optimum on 99.0% of raw reads on the native HUBO and on 0% after quadratisation (187–197 variables). The auxiliaries are tied to their pairs by penalties $M > S$, creating narrow valleys that single-flip dynamics cannot cross.

4.5 E10: penalty form (quick run, 2 instances)

Question. Do slack-free penalties help?

Setup. Unbalanced and linear penalties against the quadratic slack penalty; 10 vs 13 variables.

Output. `results/E10_penalty_form/`

Reproduce. `python scripts/run_experiments.py --only E10`

Reading. The unbalanced penalty at $0.1\Lambda_{\text{safe}}$ cuts the dynamic range from 163 to 16.5 with an optimal ground state on these instances, but its exactness is instance-dependent. The linear penalty is a Lagrangian relaxation: its ground states are feasible or optimal only for lucky multipliers. This needs a full run before any claim.

Table 4.5: E10 (generated).

form	weight	vars	dyn. range	ground feas.	ground opt.	P(opt) pp
quadratic slack (exact)	Λ_{safe}	13	163.0	1.000	1.000	0.719
unbalanced	$0.1 \cdot \Lambda_{\text{safe}}$	10	16.5	1.000	1.000	0.859
unbalanced	$0.3 \cdot \Lambda_{\text{safe}}$	10	49.0	1.000	1.000	0.766
unbalanced	$1.0 \cdot \Lambda_{\text{safe}}$	10	163.0	1.000	1.000	0.812
linear	$0.5 \cdot \mu$	10	0.6	0.000	0.000	1.000
linear	$1.0 \cdot \mu$	10	0.4	0.500	0.500	0.500
linear	$2.0 \cdot \mu$	10	1.0	1.000	0.000	1.000

4.6 E11: native HUBO versus Rosenberg QUBO spectra (quick run, 2 instances)

Question. Does quadratisation change the annealing bottleneck?

Setup. Cubic ($K=3$) objective, 6 decision variables; exact spectra.

Output. results/E11_pubo_gap/

Reproduce. `python scripts/run_experiments.py --only E11`

Table 4.6: E11 (typed from docs/RESULTS.md).

model	qubits	final gap	P(ground), $T=10$
A: native cubic HUBO + slack penalty	9	$9.4 \times 10^{-5} / 1.1 \times 10^{-5}$	3.1×10^{-3}
B: Rosenberg QUBO of A	16	$9.4 \times 10^{-5} / 1.1 \times 10^{-5}$	2.4×10^{-5}
C: native HUBO on $\mathcal{F}$ (diagonal only)	6	$2.4 \times 10^{-2} / 3.7 \times 10^{-3}$	$4.5 \times 10^{-2} / 4.1 \times 10^{-2}$

Reading. The final gap is set by the slack penalty and is identical to three digits for A and B, but B needs 7 more qubits and its short-anneal ground-state probability is about $100\times$ lower. Removing the penalty (C, diagonal-only, not hardware-realizable) enlarges the gap about $250\times$.

Chapter 5

Classical bar

5.1 E3: solver scaling (certified optima, safe penalty)

Question. How do penalty-QUBO samplers scale against certified optima?

Setup. Two families, 3 instances per size $\times$ 4 seeds $\times$ 32 reads; 1,000 sweeps for SA/SQA, SQA with $P = 16$; SA-matched gets $(P+1) \times$ more sweeps; QAOA only up to 16 qubits.

Output. results/E3_scaling/

Reproduce. `python scripts/run_experiments.py --only E3`

Table 5.1: E3: median raw success per read (generated).

family / vars	Greedy	SA	SA-matched	SQA	SQA-fixedT	Tabu	QAOA(p=4)
park / 13	0.20	0.48	1.00	1.00	1.00	1.00	0.01
park / 17	0.06	0.03	0.48	0.56	0.84	1.00	—
park / 27	0.00	0.00	0.00	0.00	0.00	1.00	—
park / 32	0.00	0.00	0.00	0.00	0.00	0.94	—
park / 46	0.00	0.00	0.00	0.00	0.00	0.66	—
mix+equity / 42	0.00	0.00	0.00	0.00	0.00	0.00	—
mix+equity / 56	0.00	0.00	0.00	0.00	0.00	0.00	—
mix+equity / 89	0.00	0.00	0.00	0.00	0.00	0.00	—

Reading. Tabu is the strongest sampler on single-intervention QUBOs and weaker on the constrained multi-choice family. The SQA-over-SA advantage at equal sweeps largely disappears at equal flips on parks (17 variables: 0.54 ± 0.07 vs 0.50 ± 0.11); on mix + equity SQA's $+0.02$ – 0.03 benefit over SA-matched costs about $2.4 \times$ the wall time. Fixed-temperature SQA collapses in feasibility on mix + equity. QAOA at $p = 4$ on 13 qubits puts 1% on the optimum. The greedy planner is optimal on 8 of 15 park instances (worst gap 0.037°C) and infeasible on 5 of 9 mix instances because it ignores equity. After post-processing, every sampler, even random, solves most park instances, so post-processed success does not discriminate there.

5.2 E7: end-to-end showcase

Question. What does a full plan look like on a larger city?

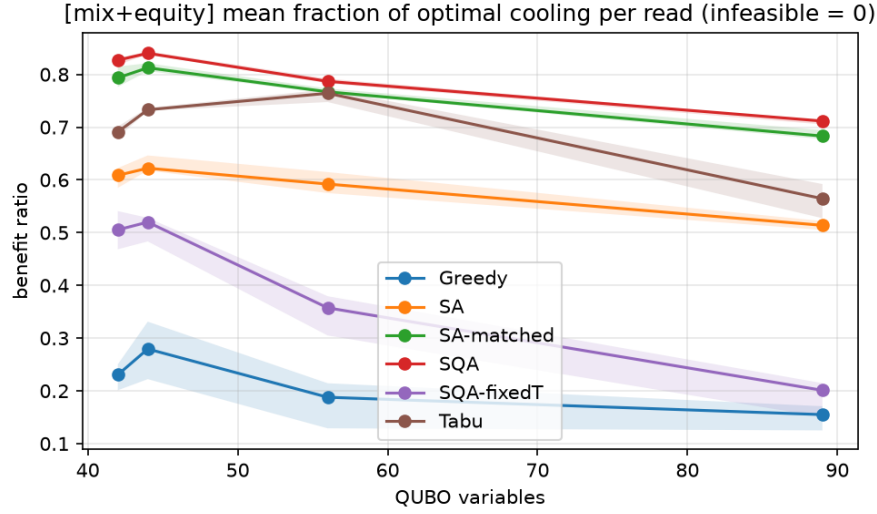


Figure 5.1: E3 mean benefit per read on mix + equity.

Setup. 14×14 city, parks, water and cool pavement, budget 24, equity; 81 decision variables and a 105-variable QUBO at $0.1 \Lambda_{\text{safe}}$.

Output. results/E7_showcase/

Reproduce. `python scripts/run_experiments.py --only E7`

Table 5.2: E7 (typed from docs/RESULTS.md).

plan	feasible	exposure T ($^{\circ}\text{C}$)	pop.-weighted cooling	interventions	time
baseline	—	32.311	—	0	—
MILP (certified optimal)	yes	31.141	1.231 $^{\circ}\text{C}$	8 parks	10.8 s
Greedy planner	yes	31.141 (same plan)	1.231 $^{\circ}\text{C}$	8 parks	< 1 s
SA + pp	yes	31.230	1.140 $^{\circ}\text{C}$	10	12.4 s
SQA + pp	yes	31.230 (same plan as SA)	1.140 $^{\circ}\text{C}$	10	32.1 s
Tabu + pp	yes	31.459	0.891 $^{\circ}\text{C}$	14	10.8 s

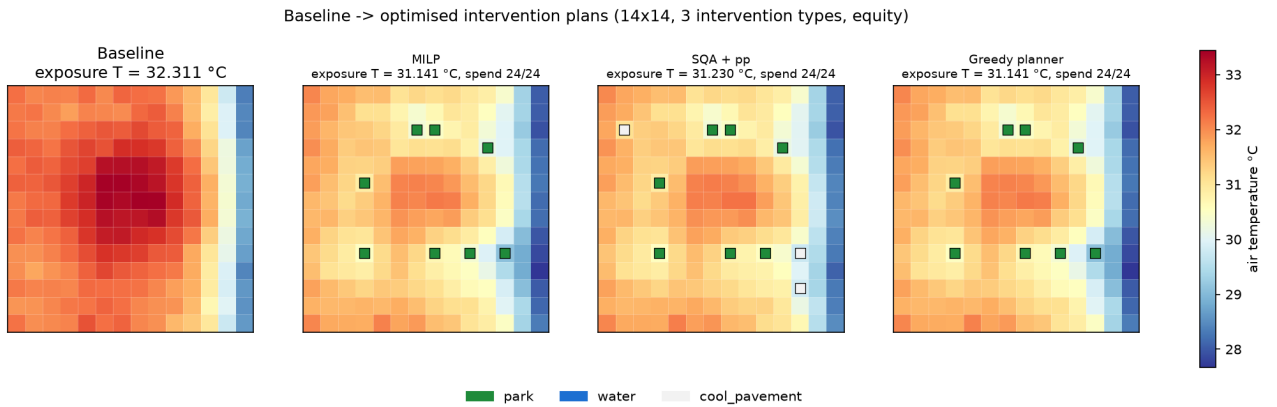


Figure 5.2: E7 baseline to optimised plans.

Reading. The optimal plan lowers the exposure-weighted temperature by 1.17°C ; all 20 baseline hot-spot cells drop below the hot-spot threshold under every plan. Water is never selected (5 units for 2.0°C is dom-

inated by parks at 3 units for 1.5°C). The hottest core receives nothing because the generator places few candidate lots in dense blocks: the siting constraint, not the optimiser, limits what can be achieved.

5.3 E9: weak penalty plus repair as a declared hybrid

Question. Is weak penalty + repair the best penalty-based recipe?

Setup. E3 mix + equity 8×8 , 10×10 , 12×12 (3 seeds each, all certified), 2 sampler seeds, $\alpha \in \{0.01, 0.03, 0.1, 1\}$.

Output. results/E9_hybrid/

Reproduce. `python scripts/run_experiments.py --only E9`

Table 5.3: E9 (generated; full table in the appendix).

solver	α	P(feas)	P(opt) raw	P(opt) pp	benefit pp
FeasibleSA	–	1.000	0.460	–	–
Greedy planner	–	0.444	0.444	–	–
MILP (HiGHS)	–	1.000	1.000	–	–
SA	0.01	0.608	0.017	0.503	0.941
SA	1	0.979	0.000	0.031	0.813
SA-matched	0.01	0.715	0.168	0.804	0.976
SA-matched	1	1.000	0.003	0.184	0.896
SQA	0.01	0.833	0.156	0.767	0.979
SQA	1	1.000	0.003	0.108	0.880
Tabu	0.01	0.792	0.078	0.321	0.903
Tabu	1	0.983	0.000	0.009	0.748

Reading. $\alpha = 0.01$ plus repair raises post-processed $P(\text{opt})$ from at most 0.18 (safe Λ) to 0.77–0.80 for SQA and SA-matched. SQA gains nothing over compute-matched SA. Greedy violates equity on 5 of 9 instances; MILP certifies in 3.8 s on average. Nothing here meets MILP’s $P(\text{opt}) = 1$.

5.4 HiGHS wall-clock across studies

study	instances	HiGHS wall
E7 showcase	1	10.8 s
E9 (mix + equity)	9	3.8 s mean
E12 family H	9	0.81 s mean, 2.29 s max
E12b family H	9	0.77 s mean
E12b family C	3	0.013 s mean
E13 tile	1	1.67 s

Certification becomes slow above about 60 dense decision variables: a 20×20 pilot was not certified in 120 s.

Chapter 6

Quantum simulation and QAOA

6.1 E6: small-scale quantum simulation (14 qubits)

Question. What limits analog annealing and QAOA on the certified penalty QUBO?

Setup. 7×7 city, 10 decision + 4 slack qubits, $H(s) = -(1-s)\sum X + sH_P$ with H_P normalised to $[0, 1]$; exact spectra, simulated anneals and QAOA.

Output. results/E6_quantum/

Reproduce. `python scripts/run_experiments.py --only E6`

Table 6.1: E6 (typed from docs/RESULTS.md).

$\Lambda/\Lambda_{\text{safe}}$	ground = opt.	$\Delta(s=1)$	$1/\Delta^2$	QAOA $P(\text{opt}), p=6$
0.1	yes	7.1×10^{-5}	2.0×10^8	0.0019
0.3	yes	2.4×10^{-5}	1.8×10^9	0.0018
1	yes	7.1×10^{-6}	2.0×10^{10}	0.0017
3	yes	2.4×10^{-6}	1.8×10^{11}	0.0017
10	yes	7.1×10^{-7}	2.0×10^{12}	0.0017

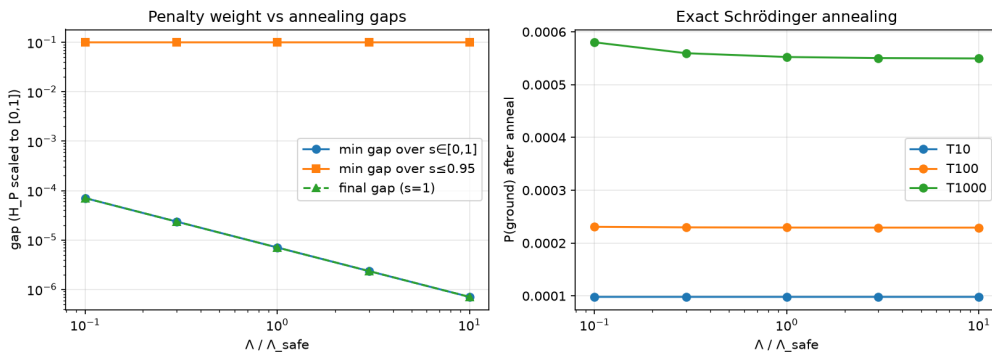


Figure 6.1: E6 gap versus penalty.

Reading. The final gap scales exactly as $1/\Lambda$ and the minimum gap sits at $s = 1$, so $1/\Delta^2$ grows as Λ^2 , from 2×10^8 to 2×10^{12} . Practical anneal times ($T \leq 1000$) give about $9 \times$ uniform ground-state probability, independent of Λ . QAOA's approximation ratio (≈ 0.998) is measured on a penalty-dominated scale; its

optimum probability only grows from 0.08% ($p = 1$) to 0.17–0.19% ($p = 6$), about $30\times$ uniform ($1/2^{14} = 6.1 \times 10^{-5}$).

In plain language. Making the rules strict makes the energy differences between good and bad plans tiny, and a quantum annealer would need impossibly long runs to tell them apart.

6.2 E8: constraint-preserving mixers ($p \leq 2$)

Question. Do Dicke/XY and feasibility-projected mixers fix penalty QAOA?

Setup. 8 park instances (including the “exactly 3” twin), 6 mix + equity instances with $|\mathcal{F}|$ from 720 to 2,243; $p \in \{1, 2\}$, one L-BFGS-B run from a linear ramp plus one random restart; heuristics 32 reads $\times$ 3 seeds.

Output. results/E8_mixers/

Reproduce. python scripts/run_experiments.py --only E8

Table 6.2: E8 (typed from docs/RESULTS.md; full table in the appendix).

family	method	$P(\text{opt})$	$P(\text{feas})$	benefit	wall (s)
park	MILP	1	1	1	0.09
park	Greedy planner	0.86	1	1.00	0.001
park	XY-QAOA (swap+add/remove) $p=1 / p=2$	0.063 / 0.197	1	0.87 / 0.94	1.6 / 7.1
park	X-mixer penalty QAOA $p=1 / p=2$	0.008 / 0.011	0.70 / 0.80	0.40 / 0.51	0.6 / 1.1
park	FeasibleSA	1	1	1	0.26
park	SA / Tabu (safe- Λ QUBO)	0.25 / 1	1	0.96 / 1	0.07 / 0.02
exactly-3	Dicke-XY complete $p=1 / p=2$	0.18 / 0.37	1	0.94 / 0.97	3 / 15
exactly-3	Dicke-XY ring $p=1 / p=2$	0.04 / 0.14	1	0.88 / 0.91	3 / 16
exactly-3	X-mixer penalty QAOA $p=1 / p=2$	0.004 / 0.005	0.45 / 0.62	0.35 / 0.49	0.2 / 0.3
mix+eq	MILP	1	1	1	0.18
mix+eq	Greedy planner	0.33	0.33	0.33	0.002
mix+eq	XY-QAOA $p=1 / p=2$	0.002 / 0.003	1	0.60 / 0.61	3.6 / 25
mix+eq	FeasibleSA	0.39	1	0.92	0.79
mix+eq	SA / Tabu (safe- Λ QUBO)	0.002 / 0	1.0	0.61 / 0.61	0.2 / 0.14

Reading. $P(\text{feasible}) = 1$ by construction, and $P(\text{opt})$ is $10\text{--}70\times$ higher than X-mixer penalty QAOA at equal depth. On mix + equity, XY-QAOA barely beats uniform sampling of $\mathcal{F}$ (≈ 0.002) while FeasibleSA reaches 0.39 per read. Complete-graph XY beats ring XY. Constraint-preserving QAOA closes the gap to penalty QAOA, not to greedy or MILP.

6.3 E8b: Dicke-XY depth curve

Question. Is $p \leq 2$ simply too shallow?

Setup. Three “exactly 3 parks” 7×7 cities (seeds 4, 5, 6), $n = 10$, $|\mathcal{F}| = 120$, a single certified optimum each; $p = 1, \dots, 6$; L-BFGS-B from a linear ramp plus one random restart, maxiter 300.

Output. results/E8b_depth/

Reproduce. `python scripts/run_experiments.py --only E8b`

Table 6.3: E8b (generated): mean [min–max] $P(\text{opt})$.

method	$p=1$		$p=2$	$p=3$	$p=4$	$p=6$
Dicke-XY complete	0.134	0.242 [0.08–0.37]	0.309	0.376	0.427 [0.17–0.65]	
Dicke-XY ring	0.034	0.067 [0.02–0.14]	0.057	0.066	0.111 [0.02–0.26]	
X-mixer penalty QAOA	0.004	0.005	0.006	0.008		0.009

method	$P(\text{opt})$	benefit	wall (s)
MILP (HiGHS)	1.000	1.000	0.317
Greedy planner	0.667	1.000	8.54×10^{-4}
uniform on F	0.008	0.849	0.000
FeasibleSA	1.000	1.000	0.206

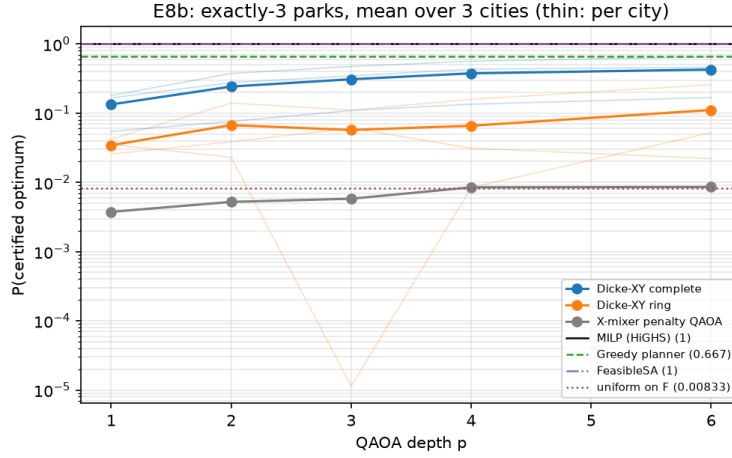


Figure 6.2: E8b $P(\text{opt})$ versus depth.

Reading. $P(\text{opt})$ grows roughly logarithmically in p (0.13 to 0.43) and is still below 0.5 at $p = 6$; the worst city reaches 0.17. Ring-XY at $p = 6$ (0.11) is below complete-XY at $p = 1$ (0.13), and its curve is not monotone, a sign that one random restart does not reliably find the global angles. The X-mixer puts 0.9991 of its mass on feasible strings at $p = 6$ but $P(\text{opt}) = 0.0086$, essentially uniform over $\mathcal{F}$: it learns the constraint, not the objective. Expected benefit for Dicke-XY complete rises from 0.955 to 0.993. Greedy misses the surrogate optimum on one city but scores 1.001 on the exact physics, a surrogate effect.

For specialists. ConstrainedQAOA is an exact state-vector simulation on $\text{span}(\mathcal{F})$ from a dense $|\mathcal{F}| \times |\mathcal{F}|$ eigendecomposition. It is not a compiled gate circuit. At $p = 6$ the wall-clock (8–15 s per city) is spent in the classical angle optimiser.

6.4 ANIM: animated comparisons

Question. How does each method move in its own search space, and what does its incumbent do to the temperature field?

Setup. Showcase A: the E8b seed-4 “exactly 3” city ($n = 10$). Showcase B: the E9 10×10 mix + equity city, seed 0 ($n = 39$, budget 18). At most 40 frames per GIF at 10 fps; fixed temperature and cooling scales

per instance.

Output. results/ANIM/ (23 GIFs, manifest.json)

Reproduce. python scripts/run_experiments.py --only ANIM

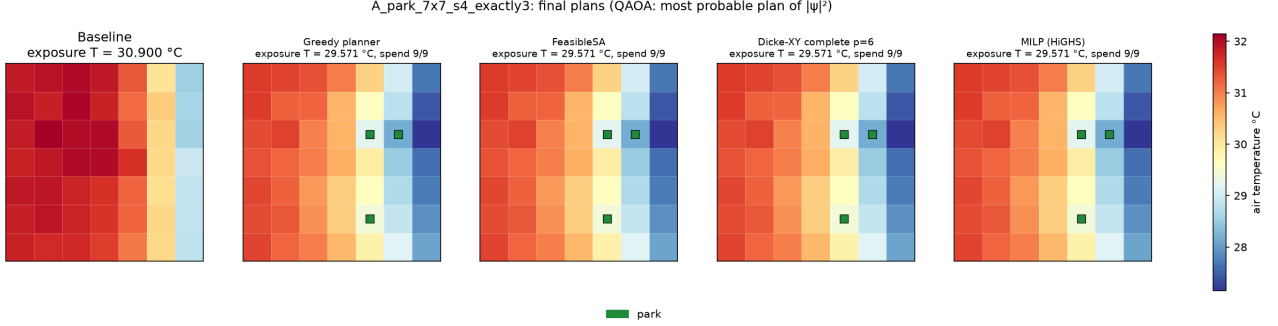


Figure 6.3: Showcase A, final plans: greedy, FeasibleSA, Dicke-XY $p = 6$ (most probable plan), MILP.

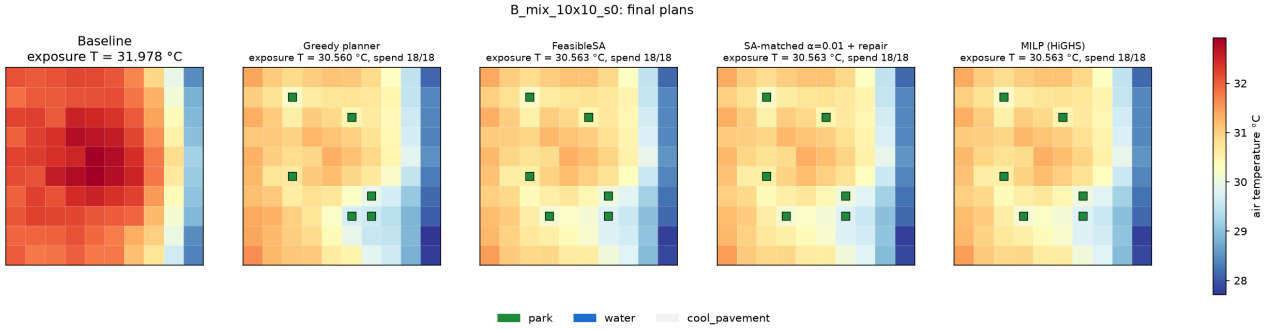


Figure 6.4: Showcase B, final plans: greedy (infeasible on equity), FeasibleSA, weak-penalty hybrid (in place of QAOA, $|\mathcal{F}| > 4000$), MILP.

Reading. In showcase A all four columns end on the same plan ($T = 29.571$ °C); the QAOA column keeps flickering because its draws miss the optimum 35% of the time. In showcase B the greedy column ends infeasible. GIFs are not embedded; they are linked:

- https://github.com/s-siyanwal/q_uhi/blob/claude/clever-gates-ezote4/results/ANIM/compare_temp_A_park_7x7_s4_exactly3.gif
- https://github.com/s-siyanwal/q_uhi/blob/claude/clever-gates-ezote4/results/ANIM/compare_temp_B_mix_10x10_s0.gif
- https://github.com/s-siyanwal/q_uhi/blob/claude/clever-gates-ezote4/results/ANIM/search_dicke_xy_complete_p6_A_park_7x7_s4_exactly3.gif
- https://github.com/s-siyanwal/q_uhi/blob/claude/clever-gates-ezote4/results/ANIM/search_xmixer_penalty_qaoa_p6_A_park_7x7_s4_exactly3.gif

For specialists. Each method lives in a different space. QAOA GIFs show $|\psi|^2$ over $\mathcal{F}$ (or over all 2^{10} strings for the X-mixer); SA, Tabu and SQA show $f + \Delta P$ on the penalty QUBO; FeasibleSA shows constrained $f(x)$. The two energy axes are not comparable. Every map is computed from a real plan.

Chapter 7

Feasible-space PIMC: E12 and E12b

7.1 FeasibleSQA

FeasibleSQA. Path-integral Monte Carlo [3] with $P = 8$ Trotter slices, each a feasible plan, samples $\pi(X) \propto \exp(-\frac{\beta}{P} \sum_k f(x^k) + J \sum_{k,i} s_i^k s_i^{k+1})$ with $J = \frac{1}{2} \ln \coth(\beta \Gamma / P)$, using the FeasibleSA moves on one slice at a time. It reproduces the exact single-qubit $\langle \sigma_z \rangle$ and is quantum-*inspired*: no annealer implements a transverse field restricted to $\mathcal{F}$ [1].

E12: matched wall-clock. Family H has 9 instances (8×8 , 10×10 , 12×12 cities, seeds 100–102) with three intervention types, budget equal to the grid side, equity, and a quartic synergy on the two larger sizes. Each method gets T^* per run, the larger of default FeasibleSA’s median wall-clock and HiGHS’s (capped at 8 s). The pre-committed COMPARABLE rule requires $P(\text{feas}) = 1$, a family-mean benefit of at least $0.95 \times$ the best classical mean, and no run over $1.5 T^*$. FeasibleSQA passes with 0.981 (Table 7.1), but FeasibleSA (0.973) and Tabu-on-F (0.976) pass the same bar, MILP reaches 1.000 in 0.81 s, and the QAOA contestants do not qualify.

E12b: the ablation. HiGHS was faster than default FeasibleSA on every instance, so E12’s T^* was FeasibleSA’s own time and there was no padding. The difference was work: FeasibleSQA rejects moves that would leave $\mathcal{F}$ before evaluating f , so in the same wall-clock it made $3.6 \times$ as many proposals (4.66 M against 1.30 M per run). Giving every method the same proposal budget W^* (4.53 M per run on average) reverses the order: FeasibleSA 0.988, FeasibleSQA 0.983, Tabu-on-F 0.977 (Table 7.2). FeasibleSQA stays within 5% of FeasibleSA (it is ahead on 1 instance, tied on 6, behind on 2) but does not beat it. E12 must not be cited as a quantum-inspired win.

7.2 E12: comparable-performance bakeoff

Question. At matched wall-clock, does any quantum or quantum-inspired method reach 0.95 of the best classical benefit on a hard family?

Setup. Family H (9 instances) and family C (3), two sampler seeds; T^* as defined above; contestants FeasibleSQA ($P = 8$, 500 sweeps, 2 reads per batch), QAOA-seeded FeasibleSA and ConstrainedQAOA $p = 2, 4$ (only if $|\mathcal{F}| \leq 4000$), X-mixer penalty QAOA (negative control, ≤ 14 qubits). The definition was committed before the run.

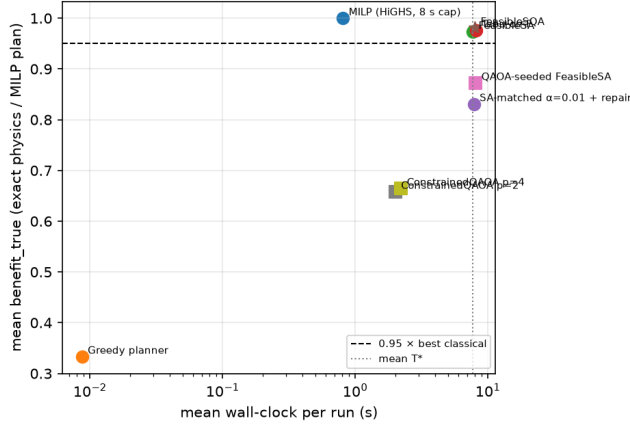
Output. results/E12_comparable/

Reproduce. python scripts/run_experiments.py --only E12

Table 7.1: E12 family H (generated; family C in the appendix).

method (family H)	P(feas)	P(opt)	benefit	min	wall (s)	comp.
MILP (HiGHS, 8 s cap)	1.000	1.000	1.000	1.000	0.809	yes
Greedy planner	0.333	0.333	0.333	0.000	0.009	no
FeasibleSA	1.000	0.778	0.973	0.813	7.693	yes
Tabu-on-F	1.000	0.778	0.976	0.708	8.120	yes
SA-matched $\alpha=0.01$ + repair	0.833	0.722	0.831	0.000	7.847	no
FeasibleSQA	1.000	0.722	0.981	0.854	7.975	yes
QAOA-seeded FeasibleSA	1.000	0.833	0.873	0.000	8.095	no
ConstrainedQAOA p=2	1.000	0.500	0.658	0.000	2.008	no
ConstrainedQAOA p=4	1.000	0.167	0.665	0.000	2.221	no

E12 family H (o classical, * quantum-inspired, s quantum sim)



E12 family C (o classical, * quantum-inspired, s quantum sim)

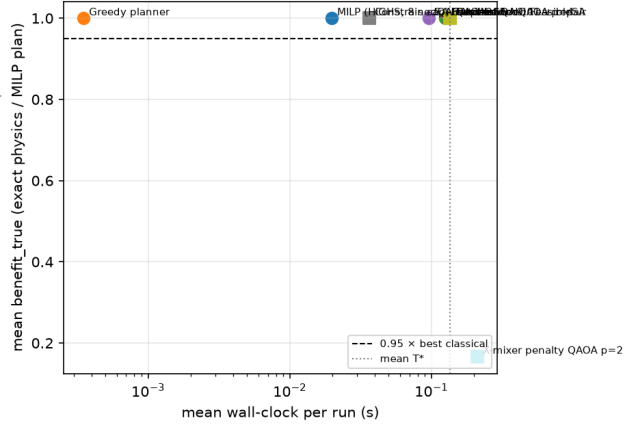


Figure 7.1: E12 benefit versus wall-clock, families H and C.

Reading. T^* is 1.4–1.6 s on 8×8 , 4.0–4.7 s on 10×10 and 18–20 s on 12×12 (0.13–0.15 s on C). MILP certifies all 12 instances. FeasibleSQA is COMPARABLE by the pre-committed rule (0.981, minimum 0.85), and so are FeasibleSA (0.973) and Tabu-on-F (0.976); the whole difference comes from the 12×12 synergy cities and is within seed noise. ConstrainedQAOA is only simulable on 8×8 ; on the $|\mathcal{F}| = 2484$ city the subspace setup exceeds T^* and scores 0. The E9 hybrid is best on 10×10 and 12×12 but fails on all three 8×8 cities because greedy repair cannot restore feasibility when the budget is tight and equity binds. On family C every method except the X-mixer control reaches 1.000.

7.3 E12b: equal-work and tight-clock ablation

Question. After removing any padding and equalising proposals, does FeasibleSQA still match FeasibleSA?

Setup. Same instances and configurations. Tight clock: FeasibleSA’s median E12 wall-clock per instance. Equal work: $W^* = \text{proposals FeasibleSQA used in E12 (median batches} \times 2 \text{ reads} \times 500 \text{ sweeps} \times 8 \text{ slices} \times n)$, 2.1–8.2 M on H and 0.3–0.5 M on C, given to every method.

Output. results/E12b-ablation/

Reproduce. `python scripts/run_experiments.py --only E12b`

Table 7.2: E12b family H, both clocks (generated).

method	clock	$P(\text{opt})$	benefit	min	wall (s)	prop. (M)	≥ 0.95 FSA	beats	w/t/l
MILP (HiGHS, 8 s cap)	none	1.000	1.000	1.000	0.77	–	–	–	–
FeasibleSA	tight	0.778	0.973	0.813	7.13	1.30	yes	no	–
Tabu-on-F	tight	0.833	0.977	0.708	7.45	2.96	yes	yes	2/5/2
FeasibleSQA	tight	0.722	0.981	0.854	7.42	4.66	yes	yes	3/6/0
QAOA-seeded FeasibleSA	tight	1.000	0.667	0.000	2.07	0.18	no	no	0/2/1
FeasibleSA	equal_work	0.889	0.988	0.854	24.99	4.53	yes	no	–
Tabu-on-F	equal_work	0.833	0.977	0.708	11.32	4.65	yes	no	1/6/2
FeasibleSQA	equal_work	0.778	0.983	0.813	7.09	4.53	yes	no	1/6/2
QAOA-seeded FeasibleSA	equal_work	1.000	1.000	1.000	2.28	2.20	yes	no	0/3/0

Reading. The tight clock reproduces E12 (FeasibleSQA 0.981, FeasibleSA 0.973) because HiGHS never set T^* . In that time FeasibleSQA made $3.6\times$ as many proposals (4.66 M vs 1.30 M). At equal proposals FeasibleSA reaches 0.988 and FeasibleSQA 0.983 (w/t/l 1/6/2): within 5% (threshold 0.938), not ahead. FeasibleSQA needs less wall-clock per proposal (7.1 s vs 25 s), an implementation property of early rejection. Tabu-on-F does not improve with more work (0.977). QAOA-seeded FeasibleSA reaches 1.000 on the three 8×8 cities with QAOA time excluded, as does FeasibleSA. Family C: every method reaches 1.000 on both clocks.

Erratum. E12’s FeasibleSQA benefit of 0.981 used $3.6\times$ the proposals of FeasibleSA. At equal proposals FeasibleSQA scores 0.983, below FeasibleSA’s 0.988. E12 must not be cited as a quantum-inspired win.

In plain language. The quantum-inspired method looked slightly better only because it tried more candidate plans in the same time. Given the same number of tries, the ordinary method is slightly better.

Chapter 8

E13: an LST-shaped tile

Question. Does a land-surface-temperature-shaped tile with scarce core lots open a classical gap?

Setup. One 16×16 city, seed 0: compact hot core ($+6^\circ\text{C}$) and cool edge (-1.5°C), population on the core, candidate lots removed from the core except 2; DEFAULT_MIX, equity, budget 16, 0.3°C synergy; $n = 108$, $|\mathcal{F}| > 4000$; tight clock 68.59 s; two sampler seeds.

Output. results/E13_tile/

Reproduce. python scripts/run_experiments.py --only E13

Table 8.1: E13 (typed from results/E13_tile/e13.md).

method	P(feas)	P(opt)	benefit	wall (s)
MILP (HiGHS, 8 s cap)	1.0	1.0	1.0000	1.67
Greedy planner	0.0	0.0	0.0000	0.06
FeasibleSA	1.0	0.5	0.9569	66.56
Tabu-on-F	1.0	0.0	0.6412	67.35
FeasibleSQA	1.0	0.5	0.9998	63.72
SA-matched $\alpha=0.01$ + repair	1.0	1.0	1.0000	64.68

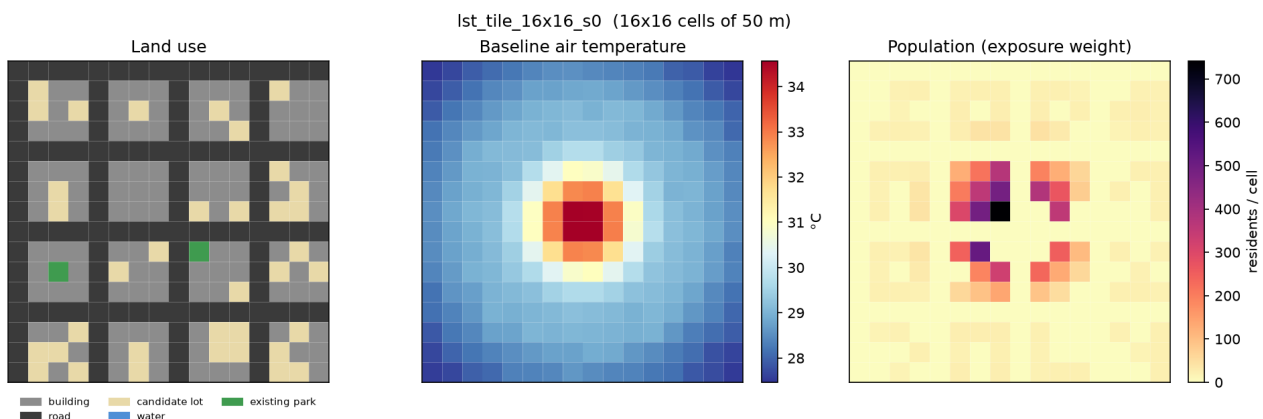


Figure 8.1: E13 tile: land use, baseline temperature, population.

Reading. E13 builds one land-surface-temperature-shaped 16×16 tile: a compact hot core with a cool edge, population concentrated on the core, and only 2 candidate lots left inside the core. With three intervention

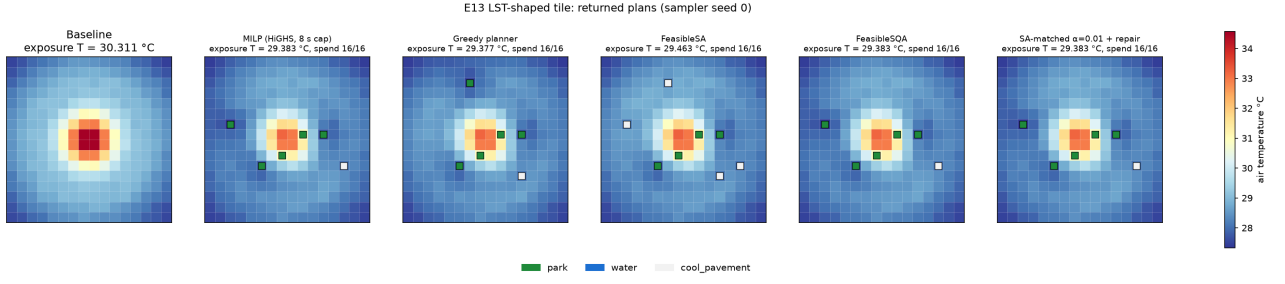


Figure 8.2: E13 returned plans (sampler seed 0).

types, equity and a budget of 16 it has $n = 108$ and $|\mathcal{F}| > 4000$, so ConstrainedQAOA cannot be simulated. At a tight clock of 68.59 s per run (Table 8.1), HiGHS certifies the tile in 1.67 s and the greedy planner is infeasible on equity, the same failure as on family H. The weak-penalty hybrid ($0.01\Lambda_{\text{safe}} + \text{repair}$) finds the optimum on both seeds, FeasibleSQA averages 0.9998 and FeasibleSA 0.9569 on two seeds each; by E12b a tight-clock comparison hands FeasibleSQA several times more proposals, so this gap is not evidence of an advantage. The mask created no classical gap.

Chapter 9

Synthesis

9.1 Hold-ups

The obstacles to a quantum or quantum-inspired result, each with the experiment that shows it:

1. **Encoding.** Slack penalties compress the objective against the spectrum width; the final gap scales as $1/\Lambda$ (E6) and is unchanged by quadratisation (E11).
2. **Rosenberg quadratisation.** $6.4\times$ as many variables and deep narrow valleys; Tabu drops from 99.0% to 0% (E5).
3. **SA matching.** SQA's advantage at equal sweeps disappears at equal flips (E3), SQA gains nothing over compute-matched SA in the hybrid (E9), and FeasibleSQA's lead disappears at equal proposals (E12b).
4. **Submodularity.** Without equity, $-f$ is monotone submodular and greedy is optimal on 8 of 15 park instances (E3) and on the E7 showcase.
5. **Mixers.** Constraint-preserving mixers fix feasibility, not optimality: $P(\text{opt}) = 0.43$ at $p = 6$ (E8b), 0.003 at $p = 2$ on mix + equity (E8).
6. **Cosmetic approximation ratios.** QAOA ratios near 0.998 on penalty-dominated scales hide optimum probabilities of 0.17–0.19% (E6).
7. **No QPU.** No constrained model was run on hardware or a hybrid cloud solver; the repository hook is inactive without a token.

9.2 Discussion

Success criterion. The criterion of Section 1 is not met by gate-model QAOA: with penalties it learns the constraint rather than the objective (E6, E8b), and with constraint-preserving mixers it remains well below FeasibleSA and MILP at every depth we simulated (E8, E8b, E12). Feasible-space PIMC meets a 0.95 bar relative to the best classical method, but it only ties feasible-space SA, and at equal proposals it is slightly behind (E12b).

What decided the outcome. Each step that improved a quantum or quantum-inspired method was a change of *encoding*: certified but gentler penalties (E1, E4, E10), native higher-order terms instead of quadratisation (E5, E11), and moves or mixers that respect the constraints (E8, E12). Each of those changes helped the classical methods at least as much. The objective's submodularity makes greedy strong wherever equity does not bind, and HiGHS certifies every benchmark instance quickly.

What hardware could still test. A fair hardware test would submit the constrained model natively, as a CQM to a hybrid solver, on the E8b “exactly 3” city (seed 4) and a family-C park city (seed 100). A safe- Λ slack QUBO should never be sent to a QPU, because its gap is set by Λ (E6, E11). The repository contains a hook that is inactive without an API token; no D-Wave result exists.

9.3 Limitations

- **Synthetic cities.** All cities and the E13 tile are synthetic; no Landsat LST or census layers were used, and intervention amplitudes and decay lengths are illustrative.
- **Surrogate physics.** The saturation law stands in for CFD or ENVI-met; MILP certifies the $K=2$ surrogate, not the exact physics.
- **PIMC $\neq$ QPU.** SQA and FeasibleSQA are classical Monte Carlo samplers [1].
- **Simulated circuits.** ConstrainedQAOA is a subspace simulation on $\text{span}(\mathcal{F})$, not a compiled circuit, and X-mixer QAOA is limited to ≤ 16 qubits.
- **No hardware numbers.** No D-Wave, Leap or other cloud result is reported.
- **Small samples.** Three instances $\times$ two seeds per E12/E12b cell; E10 and E11 are quick runs on two instances.

Chapter 10

Reference material

10.1 Reproducibility

All numbers come from seeded runs of

```
python scripts/run_experiments.py --only Ek (k ∈ {1,...,13, 8b, 12b, ANIM})
```

and the audit from `python scripts/audit_legacy.py`. The environment is recorded in `results/env.json`: Python 3.11.15, numpy 2.4.6, scipy 1.17.1 (HiGHS), numba 0.67.0, Linux x86_64, all solvers single-threaded on a 4-core container. E10 and E11 are quick runs. The tables in this document are regenerated with `python latex/tools/make_tables.py` and the PDFs with `make -C latex all`.

10.2 Glossary

QUBO quadratic unconstrained binary optimisation

HUBO higher-order unconstrained binary optimisation (degree > 2)

MILP mixed-integer linear program (solved with HiGHS)

SA simulated annealing; *FeasibleSA*: Metropolis on feasible plans only

SQA / PIMC simulated quantum annealing by path-integral Monte Carlo

FeasibleSQA PIMC whose Trotter slices are all feasible plans

QAOA quantum approximate optimisation algorithm; *ConstrainedQAOA*: Dicke/XY mixers simulated on $\text{span}(\mathcal{F})$

CQM constrained quadratic model (D-Wave Leap hybrid input format)

LST land surface temperature

UHI urban heat island

$\text{benefit}_{\text{true}}$ exact-physics cooling of a plan divided by that of the MILP plan; infeasible = 0.

$P(\text{opt})$ fraction of runs (or exact probability) that return the certified optimum.

Λ_{safe} certified sufficient penalty weight, $\Lambda > f^* - L(f)$.

$\mathcal{F}$ the feasible set of decision vectors.

T^* matched wall-clock per run in E12.

W^* equal proposal budget per run in E12b.

pp post-processing: repair plus feasible local search.

SA-matched SA with $(P+1) \times$ the sweeps, i.e. as many flip attempts as SQA.

w/t/l instances where a method's seed-mean benefit is above / equal to / below FeasibleSA's.

10.3 File map of results/

directory	contents
audit/	audit.json (legacy audit evidence A1–A7)
E1_encoding/	encoding_certification.csv, summary.json
E2_fidelity/	fidelity.csv, fidelity_summary.csv, fidelity.png, plans_L200_seed0.png
E3_scaling/	scaling_raw.csv, scaling_summary.csv, greedy_planner.csv, success/benefit/TTS plots
E4_penalty/	penalty_raw.csv, penalty_summary.csv, penalty_sweep.png
E5_hubo/	hubo_raw.csv, hubo_summary.csv
E6_quantum/	quantum_small.csv, gap_vs_penalty.png, spectrum_safe_penalty.png
E7_showcase/	summary.json, city, transition, cooling and convergence plots
E8_mixers/	raw.csv, summary.csv, summary_by_family.csv, e8.md
E8b_depth/	raw.csv, summary.csv, e8b.md, p_opt_vs_p.png
E9_hybrid/	raw.csv, summary.csv, summary_by_size.csv
E10_penalty_form/	raw.csv, summary.csv
E11_pubo_gap/	raw.csv
E12_comparable/	raw.csv, summary.csv, e12.md, benefit_vs_time.png, plan grids
E12b_ablation/	raw.csv, summary.csv, e12b.md
E13_tile/	raw.csv, e13.md, city.png, plans.png
ANIM/	23 GIFs, manifest.json, compare_final*.png
papers/	the three PDFs built from latex/
env.json	software environment and run timestamps

Appendix A

Full summary tables

Every table in this appendix is generated by `latex/tools/make_tables.py` from the committed summary CSVs (values rounded, not recomputed). Raw per-run CSVs are not reproduced.

A.1 E2

Table A.1: E2 surrogate fidelity (`results/E2_fidelity/fidelity_summary.csv`).

L (m)	order	benefit lost (%)	regret ($^{\circ}\text{C}$)	same plan	model err. ($^{\circ}\text{C}$)	terms
50	1	0.2020	0.0008	0.5000	0.0292	22
50	2	0.0000	0.0000	1.0000	0.0008	253
50	2-fit	0.0000	0.0000	1.0000	0.0003	253
50	3	0.0000	0.0000	1.0000	1.09×10^{-5}	1,793
100	1	0.0000	0.0000	1.0000	0.2680	22
100	2	0.2030	0.0023	0.2500	0.0297	253
100	2-fit	0.0241	0.0003	0.7500	0.0067	253
100	3	0.0000	0.0000	1.0000	0.0022	1,793
200	1	0.2895	0.0058	0.5000	1.1318	22
200	2	2.8138	0.0573	0.0000	0.2171	253
200	2-fit	0.2113	0.0043	0.5000	0.0247	253
200	3	0.0212	0.0004	0.7500	0.0483	1,793
300	1	0.0000	0.0000	1.0000	1.9316	22
300	2	10.2975	0.2479	0.0000	0.2807	253
300	2-fit	0.4161	0.0100	0.0000	0.0309	253
300	3	0.0000	0.0000	1.0000	0.1544	1,793

A.2 E3

Table A.2: E3 solver scaling, medians over instances $\times$ seeds (`results/E3_scaling/scaling_summary.csv`).

family	grid	solver	vars	P(succ)	pp	P(feas)	benefit	benefit pp	s/read
mix+equity	8	Greedy	42	0.000	0.031	0.359	0.239	0.802	5.01×10^{-5}
mix+equity	8	Random	42	0.000	0.031	0.000	0.000	0.809	7.68×10^{-6}
mix+equity	8	SA	42	0.000	0.062	0.984	0.617	0.837	0.004
mix+equity	8	SA-matched	42	0.000	0.266	1.000	0.805	0.905	0.074
mix+equity	8	SQA	42	0.000	0.203	1.000	0.832	0.895	0.174
mix+equity	8	SQA-fixedT	42	0.000	0.109	0.781	0.512	0.846	0.226
mix+equity	8	Tabu	42	0.000	0.000	1.000	0.703	0.770	0.004
mix+equity	10	Greedy	56	0.000	0.000	0.312	0.188	0.752	9.86×10^{-5}
mix+equity	10	Random	56	0.000	0.000	0.000	0.000	0.755	8.61×10^{-6}
mix+equity	10	SA	56	0.000	0.047	1.000	0.592	0.844	0.007
mix+equity	10	SA-matched	56	0.000	0.188	1.000	0.767	0.903	0.120
mix+equity	10	SQA	56	0.000	0.125	1.000	0.787	0.894	0.295

family	grid	solver	vars	P(succ)	pp	P(feas)	benefit	benefit pp	s/read
mix+equity	10	SQA-fixedT	56	0.000	0.062	0.609	0.357	0.840	0.395
mix+equity	10	Tabu	56	0.000	0.000	1.000	0.764	0.820	0.013
mix+equity	12	Greedy	89	0.000	0.000	0.328	0.155	0.630	2.99×10^{-4}
mix+equity	12	Random	89	0.000	0.000	0.000	0.000	0.618	1.56×10^{-5}
mix+equity	12	SA	89	0.000	0.000	1.000	0.514	0.799	0.015
mix+equity	12	SA-matched	89	0.000	0.031	1.000	0.683	0.880	0.247
mix+equity	12	SQA	89	0.000	0.016	1.000	0.712	0.851	0.615
mix+equity	12	SQA-fixedT	89	0.000	0.000	0.422	0.201	0.806	0.800
mix+equity	12	Tabu	89	0.000	0.000	0.938	0.564	0.663	0.037
park	8	Greedy	13	0.203	1.000	0.734	0.665	1.000	6.97×10^{-6}
park	8	QAOA(p=4)	13	0.012	1.000	0.785	0.524	1.000	0.010
park	8	Random	13	0.000	1.000	0.062	0.053	1.000	3.75×10^{-6}
park	8	SA	13	0.484	1.000	1.000	0.973	1.000	8.39×10^{-4}
park	8	SA-matched	13	1.000	1.000	1.000	1.000	1.000	0.014
park	8	SQA	13	1.000	1.000	1.000	1.000	1.000	0.024
park	8	SQA-fixedT	13	1.000	1.000	1.000	1.000	1.000	0.038
park	8	Tabu	13	1.000	1.000	1.000	1.000	1.000	2.04×10^{-4}
park	10	Greedy	17	0.062	1.000	1.000	0.868	1.000	1.02×10^{-5}
park	10	Random	17	0.000	1.000	0.094	0.062	1.000	4.02×10^{-6}
park	10	SA	17	0.031	1.000	1.000	0.949	1.000	0.001
park	10	SA-matched	17	0.484	1.000	1.000	0.995	1.000	0.021
park	10	SQA	17	0.562	1.000	1.000	0.995	1.000	0.037
park	10	SQA-fixedT	17	0.844	1.000	1.000	0.999	1.000	0.058
park	10	Tabu	17	1.000	1.000	1.000	1.000	1.000	5.71×10^{-4}
park	12	Greedy	27	0.000	1.000	0.469	0.418	1.000	2.15×10^{-5}
park	12	Random	27	0.000	1.000	0.016	0.008	1.000	5.30×10^{-6}
park	12	SA	27	0.000	1.000	1.000	0.909	1.000	0.002
park	12	SA-matched	27	0.000	1.000	1.000	0.963	1.000	0.042
park	12	SQA	27	0.000	1.000	1.000	0.961	1.000	0.085
park	12	SQA-fixedT	27	0.000	1.000	1.000	0.965	1.000	0.124
park	12	Tabu	27	1.000	1.000	1.000	1.000	1.000	0.002
park	14	Greedy	32	0.000	0.734	1.000	0.914	1.000	2.90×10^{-5}
park	14	Random	32	0.000	0.750	0.016	0.010	1.000	6.06×10^{-6}
park	14	SA	32	0.000	0.734	1.000	0.928	1.000	0.003
park	14	SA-matched	32	0.000	0.797	1.000	0.965	1.000	0.054
park	14	SQA	32	0.000	0.750	1.000	0.962	1.000	0.119
park	14	SQA-fixedT	32	0.000	0.719	1.000	0.966	1.000	0.165
park	14	Tabu	32	0.938	1.000	1.000	0.997	1.000	0.003
park	16	Greedy	46	0.000	1.000	1.000	0.917	1.000	6.19×10^{-5}
park	16	Random	46	0.000	1.000	0.016	0.010	1.000	8.71×10^{-6}
park	16	SA	46	0.000	1.000	1.000	0.905	1.000	0.006
park	16	SA-matched	46	0.000	1.000	1.000	0.946	1.000	0.096
park	16	SQA	46	0.000	1.000	1.000	0.945	1.000	0.242
park	16	SQA-fixedT	46	0.000	1.000	1.000	0.944	1.000	0.312
park	16	Tabu	46	0.656	1.000	1.000	1.000	1.000	0.009

A.3 E4

Table A.3: E4 penalty sweep (results/E4_penalty/penalty_summary.csv).

instance	solver	$\Lambda/\Lambda_{\text{safe}}$	P(feas)	P(succ)	pp	benefit	benefit pp
mix_8x8_equity	SA	0.010	0.104	0.010	0.417	0.097	0.780
mix_8x8_equity	SA	0.030	0.771	0.000	0.083	0.619	0.825
mix_8x8_equity	SA	0.100	0.990	0.000	0.062	0.650	0.795
mix_8x8_equity	SA	0.300	1.000	0.000	0.010	0.608	0.787
mix_8x8_equity	SA	1.000	1.000	0.000	0.031	0.599	0.773
mix_8x8_equity	SA	3.000	0.990	0.000	0.021	0.573	0.762
mix_8x8_equity	SQA	0.010	0.000	0.000	0.292	0.000	0.941
mix_8x8_equity	SQA	0.030	0.917	0.031	0.260	0.854	0.929
mix_8x8_equity	SQA	0.100	1.000	0.021	0.062	0.878	0.909
mix_8x8_equity	SQA	0.300	1.000	0.000	0.021	0.848	0.890
mix_8x8_equity	SQA	1.000	1.000	0.000	0.010	0.844	0.887
mix_8x8_equity	SQA	3.000	1.000	0.000	0.052	0.844	0.885
mix_8x8_equity	Tabu	0.010	0.198	0.010	0.469	0.188	0.826
mix_8x8_equity	Tabu	0.030	0.979	0.000	0.010	0.801	0.843
mix_8x8_equity	Tabu	0.100	1.000	0.000	0.000	0.678	0.705
mix_8x8_equity	Tabu	0.300	1.000	0.000	0.000	0.682	0.709
mix_8x8_equity	Tabu	1.000	1.000	0.000	0.000	0.682	0.709
mix_8x8_equity	Tabu	3.000	1.000	0.000	0.000	0.683	0.711

instance	solver	$\Lambda/\Lambda_{\text{safe}}$	P(feas)	P(succ)	pp	benefit	benefit pp
park_12x12	SA	0.010	0.000	0.000	1.000	0.000	1.000
park_12x12	SA	0.030	0.146	0.010	1.000	0.140	1.000
park_12x12	SA	0.100	1.000	0.000	1.000	0.914	1.000
park_12x12	SA	0.300	1.000	0.000	1.000	0.913	1.000
park_12x12	SA	1.000	1.000	0.000	1.000	0.905	1.000
park_12x12	SA	3.000	1.000	0.000	1.000	0.905	1.000
park_12x12	SQA	0.010	0.000	0.000	1.000	0.000	1.000
park_12x12	SQA	0.030	0.073	0.031	1.000	0.072	1.000
park_12x12	SQA	0.100	1.000	0.010	1.000	0.968	1.000
park_12x12	SQA	0.300	1.000	0.000	1.000	0.964	1.000
park_12x12	SQA	1.000	1.000	0.000	1.000	0.962	1.000
park_12x12	SQA	3.000	1.000	0.010	1.000	0.961	1.000
park_12x12	Tabu	0.010	0.000	0.000	1.000	0.000	1.000
park_12x12	Tabu	0.030	0.021	0.000	1.000	0.019	1.000
park_12x12	Tabu	0.100	1.000	0.990	1.000	0.999	1.000
park_12x12	Tabu	0.300	1.000	0.990	1.000	0.999	1.000
park_12x12	Tabu	1.000	1.000	0.990	1.000	0.999	1.000
park_12x12	Tabu	3.000	1.000	0.969	1.000	0.998	1.000
park_14x14	SA	0.010	1.000	0.000	0.750	0.970	1.000
park_14x14	SA	0.030	1.000	0.000	0.708	0.962	1.000
park_14x14	SA	0.100	1.000	0.000	0.750	0.954	1.000
park_14x14	SA	0.300	1.000	0.000	0.812	0.947	1.000
park_14x14	SA	1.000	1.000	0.000	0.750	0.947	1.000
park_14x14	SA	3.000	1.000	0.000	0.781	0.947	1.000
park_14x14	SQA	0.010	1.000	0.000	0.750	0.988	1.000
park_14x14	SQA	0.030	1.000	0.000	0.646	0.982	1.000
park_14x14	SQA	0.100	1.000	0.000	0.760	0.980	1.000
park_14x14	SQA	0.300	1.000	0.000	0.708	0.979	1.000
park_14x14	SQA	1.000	1.000	0.000	0.771	0.977	1.000
park_14x14	SQA	3.000	1.000	0.000	0.812	0.977	1.000
park_14x14	Tabu	0.010	1.000	0.948	1.000	1.000	1.000
park_14x14	Tabu	0.030	1.000	0.979	1.000	1.000	1.000
park_14x14	Tabu	0.100	1.000	0.906	1.000	0.997	1.000
park_14x14	Tabu	0.300	1.000	0.906	1.000	0.997	1.000
park_14x14	Tabu	1.000	1.000	0.906	1.000	0.997	1.000
park_14x14	Tabu	3.000	1.000	0.948	1.000	0.997	1.000

A.4 E5

Table A.4: E5 native HUBO vs Rosenberg QUBO (results/E5_hubo/hubo_summary.csv).

formulation	solver	vars	P(opt) raw	pp	benefit	s/read
QUBO (Rosenberg)	Greedy	192	0.000	1.000	0.849	8.72×10^{-4}
QUBO (Rosenberg)	SA	192	0.000	1.000	0.837	0.024
QUBO (Rosenberg)	SQA	192	0.000	0.997	0.930	1.115
QUBO (Rosenberg)	Tabu	192	0.000	1.000	0.847	0.029
native HUBO	Greedy	30	0.003	0.976	0.921	1.73×10^{-4}
native HUBO	SA	30	0.000	0.997	0.920	0.014
native HUBO	SQA	30	0.000	0.993	0.960	0.573
native HUBO	Tabu	30	0.990	1.000	0.999	0.179

A.5 E8

Table A.5: E8 constraint-preserving mixers by family (results/E8_mixers/summary_by_family.csv).

family	method	$P(\text{opt})$	P(feas)	benefit	wall (s)
park	MILP (HiGHS)	1.000	1.000	1.000	0.093
park	Greedy planner	0.857	1.000	1.000	0.001
park	XY-QAOA complete (swap+add/remove) p=1	0.063	1.000	0.865	1.571
park	XY-QAOA complete (swap+add/remove) p=2	0.197	1.000	0.939	7.104
park	X-mixer penalty QAOA p=1	0.008	0.703	0.395	0.623

family	method	$P(\text{opt})$	$P(\text{feas})$	benefit	wall (s)
park	X-mixer penalty QAOA p=2	0.011	0.801	0.508	1.116
park	FeasibleSA	1.000	1.000	1.000	0.263
park	SA (safe- Λ QUBO)	0.246	1.000	0.960	0.065
park	Tabu (safe- Λ QUBO)	1.000	1.000	1.000	0.023
park-cardinality	MILP (HiGHS)	1.000	1.000	1.000	0.033
park-cardinality	Greedy planner	1.000	1.000	1.000	3.44×10^{-4}
park-cardinality	Dicke-XY complete p=1	0.180	1.000	0.937	2.997
park-cardinality	Dicke-XY complete p=2	0.374	1.000	0.968	15.135
park-cardinality	Dicke-XY ring p=1	0.042	1.000	0.877	2.647
park-cardinality	Dicke-XY ring p=2	0.140	1.000	0.905	16.128
park-cardinality	X-mixer penalty QAOA p=1	0.004	0.446	0.354	0.219
park-cardinality	X-mixer penalty QAOA p=2	0.005	0.619	0.491	0.301
park-cardinality	FeasibleSA	1.000	1.000	1.000	0.243
park-cardinality	SA (safe- Λ QUBO)	0.677	1.000	0.993	0.047
park-cardinality	Tabu (safe- Λ QUBO)	1.000	1.000	1.000	0.011
mix+equity	MILP (HiGHS)	1.000	1.000	1.000	0.178
mix+equity	Greedy planner	0.333	0.333	0.333	0.001
mix+equity	XY-QAOA complete (swap+add/remove) p=1	0.002	1.000	0.599	3.572
mix+equity	XY-QAOA complete (swap+add/remove) p=2	0.003	1.000	0.612	25.157
mix+equity	FeasibleSA	0.391	1.000	0.923	0.786
mix+equity	SA (safe- Λ QUBO)	0.002	0.997	0.610	0.195
mix+equity	Tabu (safe- Λ QUBO)	0.000	1.000	0.612	0.140

A.6 E8b

Table A.6: E8b depth curve (results/E8b_depth/summary.csv); mean over 3 cities.

method	p	$P(\text{opt})$ mean	min	max	$P(\text{feas})$	benefit	wall (s)
MILP (HiGHS)	–	1.000	1.000	1.000	1.000	1.000	0.317
Greedy planner	–	0.667	0.000	1.000	1.000	1.000	8.54×10^{-4}
uniform on F	–	0.008	0.008	0.008	1.000	0.849	0.000
FeasibleSA	–	1.000	1.000	1.000	1.000	1.000	0.206
Dicke-XY complete	1	0.134	0.054	0.180	1.000	0.955	0.246
Dicke-XY complete	2	0.242	0.076	0.374	1.000	0.976	1.400
Dicke-XY complete	3	0.309	0.109	0.472	1.000	0.983	1.968
Dicke-XY complete	4	0.376	0.134	0.562	1.000	0.989	4.084
Dicke-XY complete	6	0.427	0.168	0.651	1.000	0.993	11.262
Dicke-XY ring	1	0.034	0.026	0.042	1.000	0.913	0.288
Dicke-XY ring	2	0.067	0.023	0.140	1.000	0.929	1.176
Dicke-XY ring	3	0.057	1.14×10^{-5}	0.111	1.000	0.943	2.203
Dicke-XY ring	4	0.066	0.008	0.158	1.000	0.948	3.481
Dicke-XY ring	6	0.111	0.022	0.259	1.000	0.959	11.174
X-mixer penalty QAOA	1	0.004	0.004	0.004	0.446	0.379	0.109
X-mixer penalty QAOA	2	0.005	0.005	0.005	0.619	0.526	0.077
X-mixer penalty QAOA	3	0.006	0.006	0.006	0.684	0.581	0.368
X-mixer penalty QAOA	4	0.008	0.008	0.009	0.995	0.846	0.806
X-mixer penalty QAOA	6	0.009	0.009	0.009	0.999	0.850	5.778

A.7 E9

Table A.7: E9 weak penalty + repair (results/E9_hybrid/summary.csv).

solver	α	$P(\text{feas})$	$P(\text{opt})$ raw	$P(\text{opt})$ pp	benefit pp	s/read pp
FeasibleSA	–	1.000	0.460	–	–	–
Greedy planner	–	0.444	0.444	–	–	–
MILP (HiGHS)	–	1.000	1.000	–	–	–
SA	0.01	0.608	0.017	0.503	0.941	0.071
SA	0.03	0.851	0.000	0.175	0.879	0.083
SA	0.1	0.984	0.000	0.083	0.838	0.085

solver	α	P(feas)	P(opt) raw	P(opt) pp	benefit pp	s/read pp
SA	1	0.979	0.000	0.031	0.813	0.086
SA-matched	0.01	0.715	0.168	0.804	0.976	0.253
SA-matched	0.03	0.958	0.010	0.368	0.931	0.263
SA-matched	0.1	1.000	0.000	0.241	0.910	0.258
SA-matched	1	1.000	0.003	0.184	0.896	0.249
SQA	0.01	0.833	0.156	0.767	0.979	0.535
SQA	0.03	0.969	0.010	0.323	0.921	0.535
SQA	0.1	1.000	0.000	0.125	0.892	0.529
SQA	1	1.000	0.003	0.108	0.880	0.515
Tabu	0.01	0.792	0.078	0.321	0.903	0.071
Tabu	0.03	0.967	0.000	0.019	0.796	0.077
Tabu	0.1	0.983	0.000	0.009	0.748	0.075
Tabu	1	0.983	0.000	0.009	0.748	0.077

A.8 E10

Table A.8: E10 penalty form, quick run on 2 instances (results/E10_penalty_form/summary.csv).

form	weight	vars	dyn. range	ground feas.	ground opt.	P(opt) raw	pp
quadratic slack (exact)	Λ_{safe}	13	162.952	1.000	1.000	0.344	0.719
unbalanced	$0.1 \cdot \Lambda_{\text{safe}}$	10	16.504	1.000	1.000	0.531	0.859
unbalanced	$0.3 \cdot \Lambda_{\text{safe}}$	10	49.048	1.000	1.000	0.500	0.766
unbalanced	$1.0 \cdot \Lambda_{\text{safe}}$	10	162.952	1.000	1.000	0.562	0.812
linear	$0.5 \cdot \mu$	10	0.640	0.000	0.000	0.000	1.000
linear	$1.0 \cdot \mu$	10	0.410	0.500	0.500	0.500	0.500
linear	$2.0 \cdot \mu$	10	1.035	1.000	0.000	0.000	1.000

A.9 E12

Table A.9: E12 summary (results/E12_comparable/summary.csv).

method	fam.	$ F $	P(feas)	P(opt)	benefit	min	wall (s)	T^* (s)	comp.
MILP (HiGHS, 8 s cap)	H	911,822, 20k, 2484	1.000	1.000	1.000	1.000	0.809	8.379	yes
Greedy planner	H	911,822, 20k, 2484	0.333	0.333	0.333	0.000	0.009	8.379	no
FeasibleSA	H	911,822, 20k, 2484	1.000	0.778	0.973	0.813	7.693	8.379	yes
Tabu-on-F	H	911,822, 20k, 2484	1.000	0.778	0.976	0.708	8.120	8.379	yes
SA-matched $\alpha=0.01$ + repair	H	911,822, 20k, 2484	0.833	0.722	0.831	0.000	7.847	8.379	no
FeasibleSQA	H	911,822, 20k, 2484	1.000	0.722	0.981	0.854	7.975	8.379	yes
QAOA-seeded FeasibleSA	H	911,822, 20k, 2484	1.000	0.833	0.873	0.000	8.095	8.379	no
ConstrainedQAOA p=2	H	911,822, 2484	1.000	0.500	0.658	0.000	2.008	1.463	no
ConstrainedQAOA p=4	H	911,822, 2484	1.000	0.167	0.665	0.000	2.221	1.463	no
MILP (HiGHS, 8 s cap)	C	46	1.000	1.000	1.000	1.000	0.020	0.136	yes
Greedy planner	C	46	1.000	1.000	1.000	1.000	3.49×10^{-4}	0.136	yes
FeasibleSA	C	46	1.000	1.000	1.000	1.000	0.125	0.136	yes
Tabu-on-F	C	46	1.000	1.000	1.000	1.000	0.135	0.136	yes
SA-matched $\alpha=0.01$ + repair	C	46	1.000	1.000	1.000	1.000	0.096	0.136	yes
FeasibleSQA	C	46	1.000	1.000	1.000	1.000	0.127	0.136	yes
QAOA-seeded FeasibleSA	C	46	1.000	1.000	1.000	1.000	0.135	0.136	yes
ConstrainedQAOA p=2	C	46	1.000	1.000	1.000	1.000	0.036	0.136	yes
ConstrainedQAOA p=4	C	46	1.000	1.000	1.000	1.000	0.136	0.136	yes
X-mixer penalty QAOA p=2	C	46	1.000	1.000	0.167	0.000	0.211	0.136	no

A.10 E12b

Table A.10: E12b ablation (results/E12b_ablation/summary.csv).

method	clock	fam.	inst.	P(feas)	P(opt)	benefit	min	wall	prop. (M)	FSA	≥ 0.95	beats	w/t/l
MILP (HiGHS, 8 s cap)	none	H	9	1.000	1.000	1.000	1.000	0.77	–	–	–	–	–
FeasibleSA	tight	H	9	1.000	0.778	0.973	0.813	7.13	1.30	0.973	yes	no	–
Tabu-on-F	tight	H	9	1.000	0.833	0.977	0.708	7.45	2.96	0.973	yes	yes	2/5/2
FeasibleSQA	tight	H	9	1.000	0.722	0.981	0.854	7.42	4.66	0.973	yes	yes	3/6/0
QAOA-seeded FeasibleSA	tight	H	3	1.000	1.000	0.667	0.000	2.07	0.18	1.000	no	no	0/2/1
FeasibleSA	equal_work	H	9	1.000	0.889	0.988	0.854	24.99	4.53	0.988	yes	no	–
Tabu-on-F	equal_work	H	9	1.000	0.833	0.977	0.708	11.32	4.65	0.988	yes	no	1/6/2
FeasibleSQA	equal_work	H	9	1.000	0.778	0.983	0.813	7.09	4.53	0.988	yes	no	1/6/2
QAOA-seeded FeasibleSA	equal_work	H	3	1.000	1.000	1.000	1.000	2.28	2.20	1.000	yes	no	0/3/0
MILP (HiGHS, 8 s cap)	none	C	3	1.000	1.000	1.000	1.000	0.01	–	–	–	–	–
FeasibleSA	tight	C	3	1.000	1.000	1.000	1.000	0.12	0.26	1.000	yes	no	–
Tabu-on-F	tight	C	3	1.000	1.000	1.000	1.000	0.12	0.14	1.000	yes	no	0/3/0
FeasibleSQA	tight	C	3	1.000	1.000	1.000	1.000	0.12	0.41	1.000	yes	no	0/3/0
QAOA-seeded FeasibleSA	tight	C	3	1.000	1.000	1.000	1.000	0.12	0.07	1.000	yes	no	0/3/0
FeasibleSA	equal_work	C	3	1.000	1.000	1.000	1.000	0.17	0.40	1.000	yes	no	–
Tabu-on-F	equal_work	C	3	1.000	1.000	1.000	1.000	0.32	0.40	1.000	yes	no	0/3/0
FeasibleSQA	equal_work	C	3	1.000	1.000	1.000	1.000	0.11	0.40	1.000	yes	no	0/3/0
QAOA-seeded FeasibleSA	equal_work	C	3	1.000	1.000	1.000	1.000	0.22	0.40	1.000	yes	no	0/3/0

A.11 E1, E6, E7, E11, E13

These experiments have no summary.csv; their tables are given in the main text (typed from docs/RESULTS.md or results/E13_tile/e13.md; E1 generated from encoding_certification.csv).

Appendix B

Reproducing the audit

`python scripts/audit_legacy.py` imports the *unmodified* legacy code from `UHIP.Quantum-main.zip` and writes `results/audit/audit.json`. The zip is read only for this purpose; nothing in `quhi/` imports it, and `docs/LEGACY.md` marks the zip and the root `.docx` files as provenance only. The JSON keys map to the findings as follows.

key	finding	recorded evidence
A1_ising_roundtrip	F6	forward energies match; round-trip matrix and energies do not
A2_green_layout_qubo	F1	legacy green-layout QUBO at area-budget fractions 0.3 and 0.33
A3_sqa_temperature	F3	$h = 0.5$, $\Gamma = 0.7$, $\beta = 2$, $P = 16$: exact -0.545 , ours -0.546 , legacy -0.997
A4_theorem_iv1_counterexample	F4	$\Lambda = 2.02$; penalised minimiser $(1, 1)$ infeasible, violation ² = 0.04
A5_four_cell_example	F7	constrained optimum $(1, 1, 0, 0)$, $f = -0.8$; symmetric reading selects one cell
A6_research_results	F2	legacy energies are penalty magnitudes; SQA plan all-zero (not cited)
A7_auto_penalty_underestimate	F5	$n = 10$: legacy weight 10, true range 100

Appendix C

Legacy findings in detail

F1 (critical): inequalities as equalities, in raw m^2

`core/qubo_formulation/constraint_penalty.py` implements $\sum a_i x_i \leq A_{\max}$ as $P(\sum a_i x_i - A_{\max})^2$, an equality penalty that punishes plans using less than the budget. With $a_i = 2500 \text{ m}^2$, the coefficient ratio in the legacy 4×4 green-layout problem is $|Q|_{\max} = 1.72 \times 10^9$ against $|Q_{\text{phys}}|_{\max} = 2.0 (8.6 \times 10^8)$. At an area fraction of 0.33 (3.63 cells) the global minimum selects 4 cells = $10,000 \text{ m}^2 > 9,075 \text{ m}^2$, which is infeasible. *Fix*: integer data, bounded slack bits, $\geq$ constraints with negative slack. Evidence: `audit.json` A2.

F2 (critical): shipped results are artefacts

The legacy results report penalty magnitudes as QUBO energies, an SQA energy of 0.0 with an all-zero plan, and a “best algorithm” chosen on differences in the eighth significant digit. They say nothing about UHI mitigation or quantum versus classical methods and are not cited. Evidence: A6.

F3 (critical): SQA at the wrong temperature

The legacy action uses $\sum_k E(s^k) - J_\tau \sum ss$ with acceptance $\exp(-\beta \Delta E)$; the Suzuki–Trotter action has β/P on each slice’s problem term. The legacy sampler is P times too cold with a transverse field effectively P times too weak: $\langle \sigma_z \rangle = -0.997$ against the exact -0.545 . Evidence: A3; unit test `tests/test_solvers.py`.

F4 (major): Theorem IV.1 false as stated

$\Lambda > f_{\max} - f_{\min}$ needs every violation to satisfy $g(x)^2 \geq 1$, which holds only for integer data. Counter-example: minimise $-x_0 - x_1$ subject to $0.6x_0 + 0.6x_1 \leq 1$ with $\Lambda = 2.02$; the penalised minimiser $(1, 1)$ is infeasible (violation² = 0.04). The corrected statement, $\Lambda > f^* - L(f)$ with integer data and slack bits, is proved in `docs/BACKGROUND.md` §2 and certified by E1. Evidence: A4.

F5 (major): automatic penalty weight is not a bound

The legacy estimate $\sum |\lambda_i(Q)|$ of the range of $x^T Q x$ can be too small by a factor n : $Q = \mathbf{1}$, $n = 10$ gives 10 while $\max x^T Q x = 100$. Evidence: A7.

F6 (major): broken QUBO $\leftrightarrow$ Ising round trip

Off-diagonals are built as $-4J_{ij}$ on both triangles of a symmetric Q , doubling every coupling: $\begin{pmatrix} 1 & -2 & 0 \\ -2 & 3 & -1 \\ 0 & -1 & 2 \end{pmatrix}$

returns as $\begin{pmatrix} 1 & -4 & 0 \\ -4 & 3 & -2 \\ 0 & -2 & 2 \end{pmatrix}$. quhi uses one canonical sparse polynomial and exports upper-triangular Q only.

Evidence: A1.

F7 (major): wrong worked examples and known optima

A symmetric example matrix read as $x^T Q x$ doubles couplings and selects one cell where two are required; Theorem II.3's proof sets $Q_{ii} = c_i/2$ instead of c_i ; and a 4-cell benchmark claims the optimum $[0, 1, 1, 0]$ with energy -19.25 , whereas $f([0, 1, 1, 0]) = -5$ and $[1, 0, 1, 0]$ gives -6 . Evidence: A5.

F8: other issues

See Chapter 2: inconsistent sign of the pairwise physics, 19 failing legacy tests, qiskit-only QAOA/VQE modules, missing seeds, and a scaling argument that does not transfer.

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